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Farrerol exhibits inhibitory effects on lung adenocarcinoma cells by activating the mitochondrial apoptotic pathway

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Abstract

Farrerol is an herbal compound extracted from rhododendron. Here, our study is to investigate biological effects of farrerol on lung adenocarcinoma (LAC) cells. Human LAC cell lines and xenograft mouse model were utilized to define the effects of farrerol on tumor growth. Our findings indicated that farrerol significantly reduced LAC cell viability as well as the colony-forming capacity. Flow cytometry analysis demonstrated that farrerol contributed to cell apoptosis and G0/G1 phase cell cycle arrest. Mechanistically, farrerol treatment upregulated proapoptotic molecules (Bak, Bid, cleaved caspase-3 and cleaved caspase-9) and senescence markers (p16 and p2), but downregulated antiapoptosis genes (Bcl-2 and Bcl-XL) and cell cycle-associated genes (CyclinD1 and CDK4); meanwhile, the phosphorylation of retinoblastoma (Rb) protein was attenuated upon pretreatment of LAC cells with farrerol in comparison to untreated control. Further studies indicated that farrerol elevated reactive oxygen species levels, activating mitochondrial apoptotic pathway and causing cell apoptosis. However, exposure to farrerol did not result in significant apoptosis in normal lung epithelial cells, suggesting a tumor-specific effect of farrerol on LAC cells. In animal model, farrerol showed a significant inhibitory effect on LAC xenograft tumor growth. And gene expressions in tumor tissues, as mentioned above, were in line with the in vitro results. Taken together, these results suggested that farrerol caused LAC cell apoptosis by activating mitochondrial apoptotic pathway, whereas farrerol treatment had no notable effect on normal lung epithelial cells. Farrerol might be an effective therapeutic drug for LAC.

KEYWORDS

apoptosis, farrerol, lung adenocarcinoma, mitochondrial apoptosis pathway, reactive oxygen species

1 | INTRODUCTION

Lung cancer (LC) is a common malignancy with the highest morbidity and mortality worldwide, killing hundreds of thousands of patients annually.[1,2] Nonsmall-cell lung cancer (NSCLC) accounts for approximately four-fifths of all cases, among which lung adenocarcinoma (LAC) accounts for the majority of NSCLC (40–50%).[3,4] Although various emerging therapies have been applied in recent years, chemotherapy remains one of the most common treatments for LC patients.[5] The strong side effects of existing chemotherapeutic drugs

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lead to poor compliance and acquired chemoresistance is a great challenge during cancer therapy. As a result, the cure and survival rates of NSCLC remain very low.[6,7] Hence, it is of significance to search for effective therapeutic agents with fewer side effects.

Traditional Chinese medicine has a unique efficacy in treating NSCLC and plays a crucial role in improving prognosis of NSCLC patients.[8–10] Farrerol is a dihydroflavonoid isolated from herbal medicine[11,12] and has been widely used as anti-inflammatory drug.[13–15] Besides, farrerol can be artificially synthesized, with few side effects and excellent prospects for utilization. Farrerol shares the same core structure with naringenin, hesperidin, and eriodictyol. Previous evidence has indicated that hesperidin induces cell cycle arrest and apoptosis in NSCLC, suppressing proliferative, migratory, and invasive capabilities of lung adenocarcinoma cells.[8,9] Naringenin has proven to possess multiple antitumor activities, inhibiting malignant phenotypes of lung adenocarcinoma cells through various mechanisms.[16,17] Recent studies have reported that farrerol attenuated malignant phenotypes of lung squamous carcinoma cells via suppression of epithelial-mesenchymal transition (EMT) process,[18] in addition to its various biological functions, including antioxidant, anti-inflammatory, antibacterial, and antiangiogenic activities.[19–21] However, the anti-lung adenocarcinoma activity of farrerol still needs further exploration.

In our preliminary studies, farrerol was found to induce apoptosis in lung adenocarcinoma cells, thereby inhibiting cell proliferation. Hence, we supposed that farrerol may has therapeutic potential for patients with lung adenocarcinoma. To further investigate the specific molecular mechanisms by which farrerol suppresses the proliferation of lung adenocarcinoma cells, we implemented the present study, providing theoretical basis for clinical application of farrerol.

2 | MATERIALS AND METHODS

2.1 | Cell culture

A549 and H1975 human lung adenocarcinoma cells, together with BEAS-2B human lung epithelial cells, were obtained from the Shanghai Cell Bank of the Chinese Academy of Sciences (China), followed by subcultured and cryopreserved in our laboratory. A549 and H1975 cells were cultivated in RPMI-1640 medium (Gibco) containing 10% fetal bovine serum (FBS, Gibco), 100 U/ml penicillin/streptomycin (P/S, Gibco). BEAS-2B cells were maintained in Dulbecco's modified Eagle medium (Gibco) with a mixture of 10% FBS and 1% P/S. All cells were kept in an incubator at 37°C with 5% CO_2.

2.2 | Experimental grouping

Farrerol was purchased from Selleck Chemicals. Farrerol stock solution (20 mmol/L) was prepared by dimethyl sulfoxide (DMSO, Sigma-Aldrich) and diluted with complete growth medium to the final concentration of 10, 30, or 50 µmol/L for each experiment. Cell proliferative ability was tested by Cell Counting Kit-8 (CCK-8) assay, of which 0.25% DMSO was used as solvent control, and culture medium served as negative control. For animal experiments, solvent control and farrerol treatment groups (20 and 50 mg/kg) were set up.

2.3 | Cell proliferation assay

CCK-8 Kit (Dojindo Laboratories) was used to test cell viability. The kit uses a water-soluble tetrazolium salt to quantify the number of live cells by producing an orange formazan dye upon bio-reduction in the presence of an electron carrier. The amount of formazan is measured by absorbance at 450 nm. LAC and BEAS-2B cells were plated in 96-well dishes at a density of 1.0×10^5 cells/well for 24 h. Six replicate wells were set up. Blank control and solvent control were also set up and measured. The cells were subjected to different concentrations of farrerol. Thereafter, 10 µl of CCK-8 reagent was added into each well at desired time points following incubation for 2 h at 37°C in the dark. Absorbance at 450 nm was monitored using a microplate reader (PerkinElmer).

2.4 | Colony formation assay

LAC cells (1×10^3 cells/well) were seeded into 6-well dishes and then treated with different concentrations of farrerol, respectively, with three replicate wells per group. Culture liquid was replaced every 3 days. After 2 weeks of continuous incubation at 37°C, the cells were fixed with 4% paraformaldehyde for 20 min at room temperature (RT) and stained with 0.1% crystal violet solution (Jiangsu KeyGEN BioTECH Co. Ltd.) for 30 min at RT. Next, the plates were washed three times with phosphate-buffered saline (PBS) and air-dried. Next, images were photographed under light microscope (Nikon) and clones in each well with 50 or >50 cells were counted.

2.5 | Cell apoptosis assay

LAC cells were treated with different concentrations of farrerol. Subsequently, the cells were digested with ethylenediaminetetraacetic acid-free trypsin (KeyGEN BioTECH) and collected after centrifugation at 180*g* for 4 min. Then, the cells were gently washed three times by 1 ml of PBS and resuspended in 500 µl of 1× binding buffer. Thereafter, Annexin V-FITC and propidium iodide (PI) were added following incubation for 30 min at 4°C in the dark. Apoptotic cells were tested by flow cytometry (Beckman Coulter). Results were analyzed using FlowJo 7.6 software (TreeStar Inc).

2.6 | Cell cycle assay

LAC treated with different concentrations of farrerol were fixed in precooled 70% ethanol at 4°C overnight. The immobilized cells were resuspended in PBS and incubated with RNase A (0.25 mg/ml) for 30 min at 37°C, after which the cells were stained with PI solution (KeyGEN BioTECH) for 30min at RT in the dark. Cell cycle was evaluated by flow cytometry after filtration through a 200-mesh sieve. Data was analyzed using FlowJo 7.6 software (Treestar).

2.7 | Reactive oxygen species (ROS)

ROS levels were assessed by fluorescent probe DCFH-DA (Sigma) following the manufacturer's instructions. LAC cells treated with different concentrations of farrerol were collected and resuspended with PBS. Then, incubated with DCFH-DA at 37°C for 60 min and mixed thoroughly every 10 min. Unbound DCFH-DA was removed by washing the cells three times with PBS. Flow cytometry (excitation wavelength at 488 nm, emission wavelength at 525 nm) was carried out to measure intracellular ROS levels. Results were analyzed using FlowJo 7.6.

2.8 | JC-1 mitochondrial membrane potential measurement

LAC cells treated with farrerol for 48 h were incubated with JC-1 staining solution (Sigma-Aldrich) at 37°C for 20 min. After collection, mitochondrial membrane potential in LAC cells was evaluated using flow cytometry. Fluorescence shifts from green (emission wavelength 520 nm) to red (emission wavelength 590 nm), which is indicated as increased mitochondrial membrane potential. Data were evaluated with FlowJo 7.6 software.

2.9 | Quantitative PCR (qPCR)

Total RNA from each group was extracted by TRIzol reagent (Life Technologies) and then reverse-transcribed to complementary DNA using RT Master Mix (RR036A, Takara Biomedical Technology Co., Ltd.) when the samples were passed quality and concentration testing. Next, One Step TB Green® PrimeScript™ RT-PCR Kit (Cat#RR066A; Takara) was utilized to test the relative gene expressions. qPCR reaction conditions were as described below. Predenaturation at 95°C for 3min was followed by 40 cycles of denaturation at 95°C for 30 s, annealing at 60°C for 30 s, and extension at 72°C for 15 s. Data were then analyzed by the $2^{-\triangle\triangle ct}$ method[22] and gene expressions were normalized to the house-keeping gene GAPDH. All primer sequences are listed in Table S1.

2.10 | Western blot analysis

LAC cells or cancer tissues were lysed by adding 100 µl of precooled protein lysis buffer with 1% protease inhibitor (KeyGEN BioTECH) and placed on ice for 15 min. After measurement of protein concentration by bicinchoninic acid method (KeyGEN BioTECH), protein samples were mixed with 5× loading buffer, followed by boiling at 100°C for 7 min and flash-cooling on ice. Proteins (30 µg per lane) was separated by electrophoresis using 10% sodium dodecyl sulfate–polyacrylamide gel electrophoresis gel and then transferred onto polyvinylidene difluoride membrane preactivated with methanol. The membrane was blocked in 5% skimmed milk for 2 h at RT before incubation with primary antibodies overnight at 4°C. After washing three times with TBST, the membrane was incubated with secondary antibody for 2 h at RT. Immunoblots were visualized using the ECL chemiluminescent substrate (GE Healthcare), and results were calculated using ImageJ software (NIH). β-Actin was used as the loading control.

2.11 | Animal experiments

Eighteen 8-week-old female nude mice were obtained from Shanghai SLAC Laboratory Animal Co., Ltd., and maintained under SPF conditions. Mice were acclimatized for more than 1 week before study. In addition, to prepare single-cell suspension, A549 cells at the logarithmic phase of growth were collected by trypsin digestion, and cell density was adjusted to 1×10^7 cells/ml. Next, mice were inoculated subcutaneously into the left hind limb with 0.2 ml of A549 cell suspension (2×10^6 cells). On the second day of implantation, mice were randomly divided into three groups ($n = 6$ mice in each group). During the next 42 consecutive days, mice from the experimental groups were administrated with 20 mg/kg or 50 mg/kg of farrerol by gavage every 3 days, and the control group was given an equal volume of saline. Tumor diameter was monitored weekly with a vernier caliper and tumor volume was quantified using the formula: 1/2 (length/width^2). Mice were killed under deep anesthesia after 42 days. Subcutaneous tumor nodules were gently peeled off, photographed, and weighed. A part of tumor tissue was fixed in 10% neutral formalin; the other part was used for extracting total RNA or proteins. All animal experiments in this study were approved by the Ethics Committee of The First Affiliated Hospital of Fujian Medical University. Animal experimental procedures strictly followed the Guidelines of the China Animal Welfare Legislation for Care and Use of Laboratory Animals.

2.12 | Data analysis

Data were analyzed by IBM SPSS Statistics V22.0 (SPSS Inc.) and plotted with Graph Pad Prism 8 software (Graph Pad software). All experiments were repeated independently three times. Data were demonstrated as mean ± standard deviation. Comparisons among multiple groups were performed by one-way analysis of variance with Dunnett posttest. A $p < 0.05$ indicates that the difference is statistically significant.

3 | RESULTS

3.1 | Farrerol inhibits proliferation of lung adenocarcinoma cells

To investigate farrerol activity (Figure 1A) on lung adenocarcinoma cells and normal lung epithelial cells, we performed CCK-8 and colony formation assays. Our findings showed that the half-maximal inhibitory concentration (IC_{50}) value at 48 h for farrerol is 18.9 μM in A549 cells (Figure S1). Farrerol restrained A549 and H1975 cell proliferation in a dose-dependent manner (Figure 1B,C). After treating normal lung epithelial cells BEAS-2B with the same concentration of farrerol, we found that farrerol had no significant effect on proliferation of BEAS-2B cells; however, 200 μmol/L farrerol attenuated proliferative capability of BEAS-2B cells, indicating that farrerol has a certain degree of selectivity in killing tumor cells (Figure 1D). Besides, 0.25% DMSO as a solvent to dissolve farrerol did not affect the proliferative ability of A549 and H1975 cells. All cells treated with Cisplatin (DDP, 1 μg/ml) served as positive control. Remarkably, DDP has been shown to effectively kill tumor cells. Colony formation assay further revealed that farrerol treatment was responsible for the decreased clone numbers of A549 and H1975 cells compared with the paired untreated cells (Figure 1E,F). There was no statistical difference in the number of clones between negative control and the solvent control.

3.2 | Farrerol induces G0/G1 cell cycle arrest in lung adenocarcinoma cells

Given that farrerol treatment inhibited proliferation of lung adenocarcinoma cells and cell proliferation is coupled with cell cycle progression, we next examined the biological activity of farrerol on cell cycle distribution. We found that treatment A549 and H1975 cells with farrerol for 48 h or 72 h caused G0/G1 cell cycle arrest in a dose-dependent manner compared with the paired untreated cells (Figure 2A–D). Nevertheless, treatment of A549 and H1975 cells with farrerol for 24 h at concentrations of either 10, 30, or 50 μmol/L had no effect on cell cycle progression (Figure 2A and 2D). These results confirmed that farrerol induced G0/G1 cell cycle arrest in time- and dose-dependent manners. For better efficiency, it required a longer duration of action and treatment concentration.

Next, to understand how farrerol acts on cell cycle progression, we implemented qPCR and Western blot assays. Data demonstrated that messenger RNA (mRNA) and protein levels of CDK4 and CyclinD1, two key genes in the transition from G1 to S phase, were significantly downregulated after farrerol treatment of A549 and H1975 cells for 72 h in comparison to the paired untreated cells, but no significant changes were observed in CDK6 expression. Besides, mRNA and total protein expression of retinoblastoma tumor suppressor protein (Rb), the main target of cyclinD-CDK4/6 pathway,[23] were not significantly changed, while Rb phosphorylation was markedly attenuated in farrerol-treated A549 and H1975 cells compared to the paired untreated cells. In contrast, the mRNA and protein levels of cellular senescence markers P21 and P16 were dramatically upregulated after farrerol treatment (Figure 3A–D). These findings suggested that farrerol reduced the expression of cell-cycle-promoting genes (CDK4, CyclinD1), and elevated the expression of cell cycle-inhibitory genes (P21, P16), resulting in G0/G1 cell cycle arrest. However, these changes were not observed in A549 and H1975 cells that were subjected to farrerol for 24 h (Figure 3E), suggesting that farrerol acting on cell cycle progression was partly responsible for the inhibition of LAC cell proliferation.

3.3 | Farrerol facilitates lung adenocarcinoma cell apoptosis in time- and dose-dependent manners

During the study, we observed the presence of cell fragmentation and cell shrinkage after farrerol treatment compared with the untreated cells under optical microscopy, implying that farrerol may induce apoptosis in lung adenocarcinoma cells (Figure 4A). To explore the mechanisms by which farrerol restrained LAC cell proliferation, we performed flow cytometry to measure apoptotic cells using Annexin V-FITC (AV)/PI staining. Data confirmed that farrerol contributed to an increase of apoptotic cells in time- and dose-dependent manners in both A549 and H1975 cells compared to the matched control (Figure 4B,C). These findings suggested that farrerol promoted LAC cell apoptosis.

To unveil how farrerol regulates cell apoptosis, we implemented qPCR and Western blot assays. Results indicated that cleaved caspase3 and cleaved caspase9, two proapoptotic proteins, were significantly upregulated in farrerol-treated A549 cells compared with the untreated cells, whereas their mRNA levels were not altered. Bcl-2 and Bcl-XL, two antiapoptotic genes, were significantly downregulated, while mRNA and protein levels of Bak and Bid, two proapoptotic genes, were dramatically elevated. Besides, the mRNA and protein levels of Bax and Bad, two canonical factors involved in mitochondrial apoptotic pathway, did not change significantly; however, their phosphorylation levels were markedly reduced after farrerol treatment (Figure 5A–C). Similarly, mRNA levels of the above-mentioned genes including Bcl-2, Bcl-XL, Bak, and Bid were markedly altered in farrerol-treated H1974 cells (Figure 5D). Given that the decrease of Bax and Bad phosphorylation leads to the alteration of mitochondrial membrane permeability,[24–26] we thus supposed that farrerol-induced apoptosis was caused by activating mitochondrial apoptotic pathway.

3.4 | Farrerol treatment results in increased ROS levels and decreased mitochondrial membrane potential in lung adenocarcinoma cells

To confirm the relationship between farrerol and mitochondrial apoptosis pathway, mitochondrial membrane potential was assessed by JC-1 staining. We observed that farrerol treatment significantly

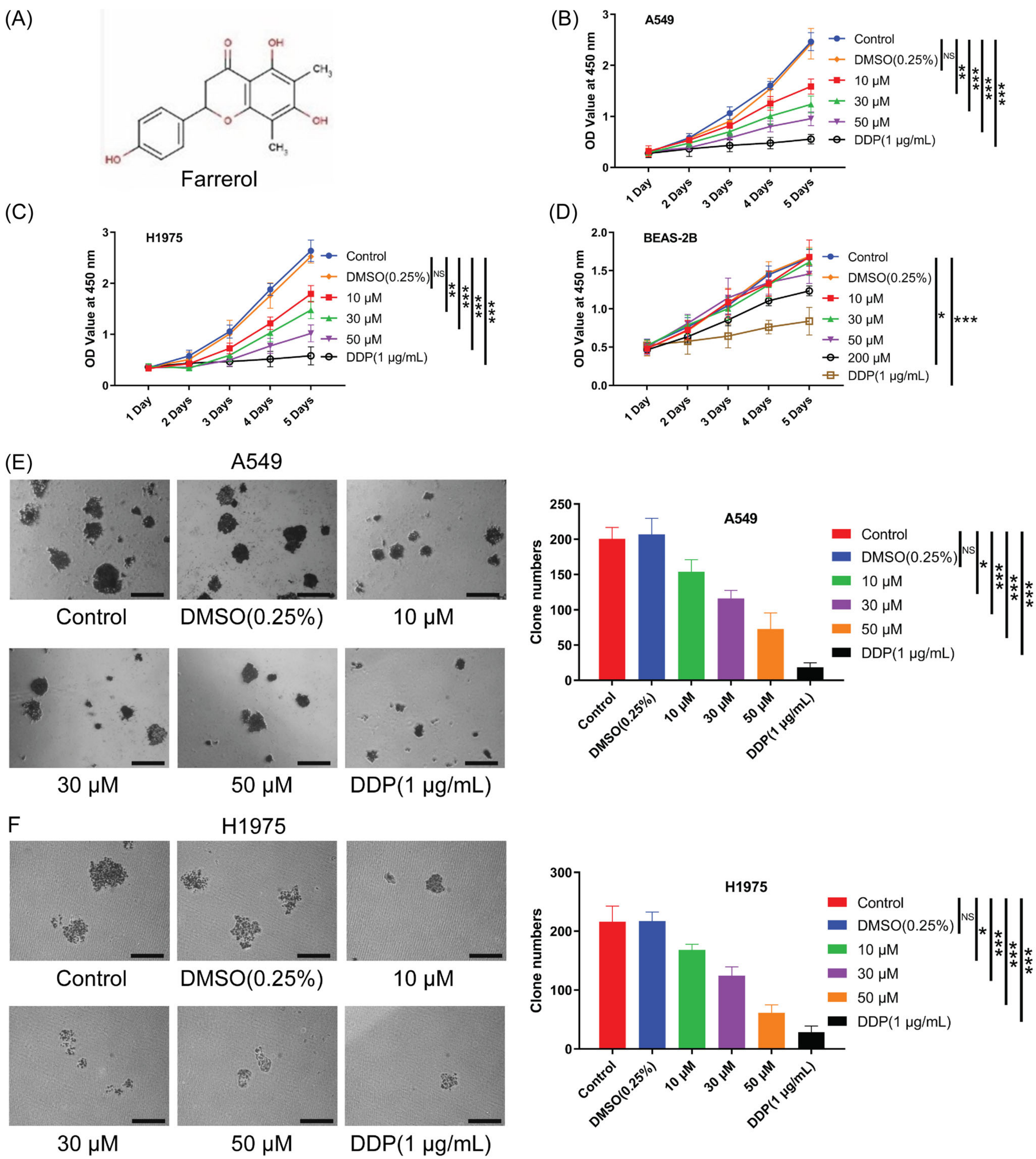

FIGURE 1 Effect of farrerol on cell viability of human lung adenocarcinoma cells and normal lung epithelial cells. (A) Molecular structure of farrerol. (B–D) CCK-8 assay was conducted to test the effect of rhododendron on the proliferation of (B) A549 and (C) H1975 cells and (D) normal lung epithelial cells. (E–F) Colony formation assay was performed to test the effect of rhododendron on the colony-forming ability of (E) A549 and (F) H1975 cells. Left panel in (E) and (F): representative images; right panel in (E) and (F): quantification. DDP treatment served as the positive control. Scale bar, 200 μm. Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, when compared with the control group. CCK-8, Cell Counting Kit-8; DDP, cisplatin; PI, propidium iodide; NS, not significant

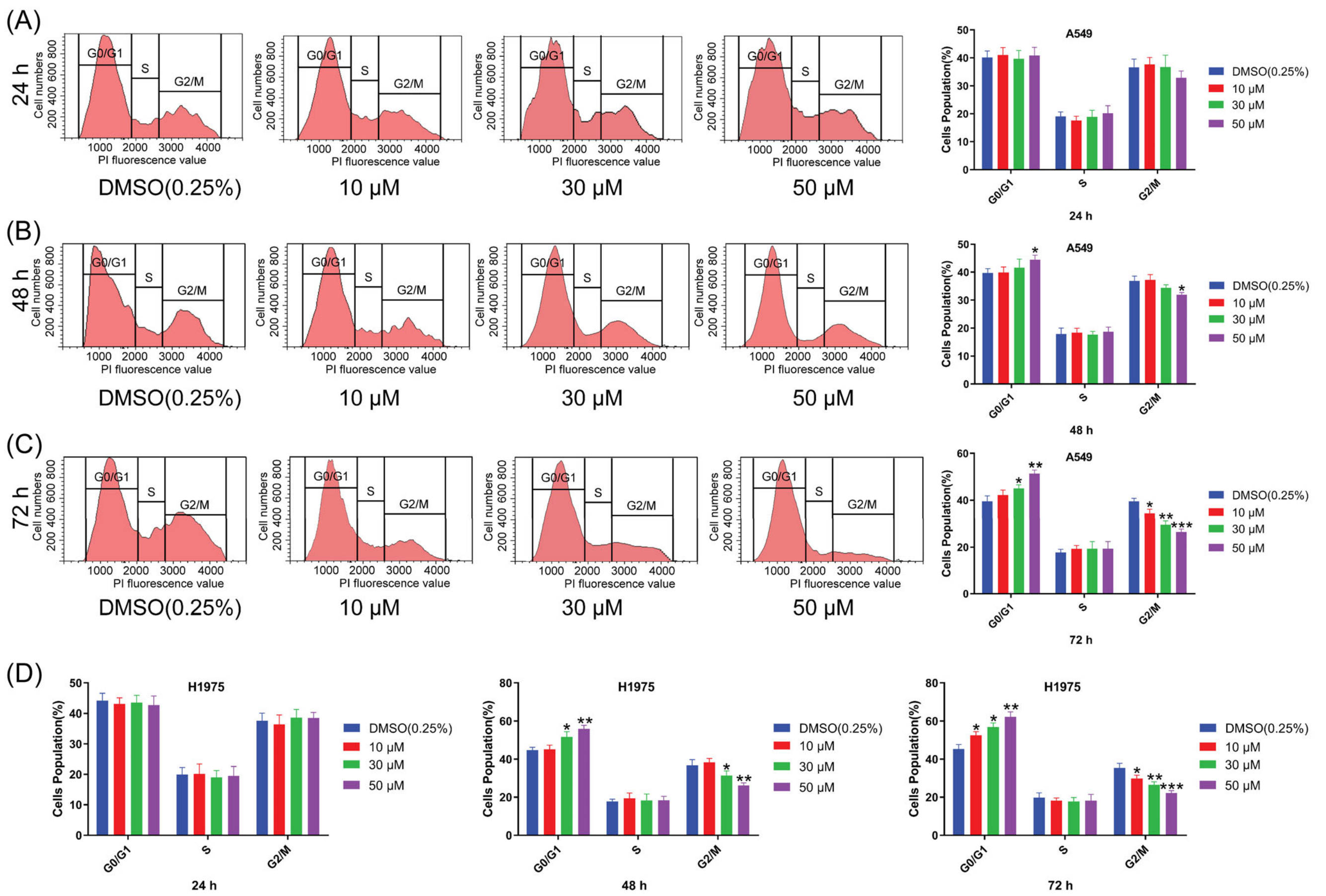

FIGURE 2 Effect of farrerol on cell cycle of human lung adenocarcinoma cells. (A, B) PI staining was used to test different concentrations of farrerol treatment for (A) 24 h, (B) 48 h, or (C) 72 h on cell cycle distribution of A549 cells. Left panel: representative images; right panel: quantification. (D) PI staining was performed to test the effect of farrerol treatment for 24 h (left), 48 h (middle), or 72 h (right) on cell cycle distribution of H1975 cells. Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, when compared with the control group. DDP, cisplatin; PI, propidium iodide; NS, not significant

reduced the mitochondrial red fluorescence in a dose-dependent manner in A549 and H1975 cells, suggesting a decrease in mitochondrial membrane potential. Conversely, the paired untreated cells (control) emitted a strong red fluorescence (Figure 6A,B). Previous studies have proved that the decrease in mitochondrial membrane potential is usually accompanied by an increase in ROS, and mitochondria are the most sensitive to ROS.[27,28] Therefore, ROS levels in A549 and H1975 cells treated with farrerol were assessed. Results showed that farrerol significantly increased ROS content in LAC cells in a dose-dependent manner compared with the control (Figure 6C,D).

Nevertheless, it is not clear whether the increase in ROS levels after farrerol treatment is a cause or a consequence of the decrease in mitochondrial membrane potential. Hence, N-acetylcysteine (NAC, 1 mM), an inhibitor of ROS, was adopted. Cotreatment with NAC and farrerol significantly reduced intracellular ROS content compared with A549 or H1975 cells treated with farrerol alone; meanwhile, the mitochondrial membrane potential and cell apoptosis rate were partially reversed by NAC in A549 or H1975 cells (Figure 7A–D). These data indicated that farrerol treatment contributed to increased ROS levels, thereby decreasing mitochondrial membrane potential in lung adenocarcinoma cells.

3.5 | Farrerol suppresses xenograft tumor growth

We next examined whether farrerol treatment can be effective for suppression of tumor growth in vivo. Nude mice were inoculated subcutaneously with A549 cells, and tumor length and width were measured weekly for 6 weeks. Our findings demonstrated that farrerol inhibited the growth of xenograft tumors in a dose-dependent manner. Forty-two days later, tumor nodules were carefully peeled off and weighed. Data showed that farrerol reduced the weight of xenograft tumors in a dose-dependent manner compared to the control group (Figure 8A,B). Besides, the mRNA and protein levels of apoptosis- and cell cycle-related genes in xenograft tumor tissues were measured. These in vivo results were in consistence with the in vitro experiments (Figure 8C,D).

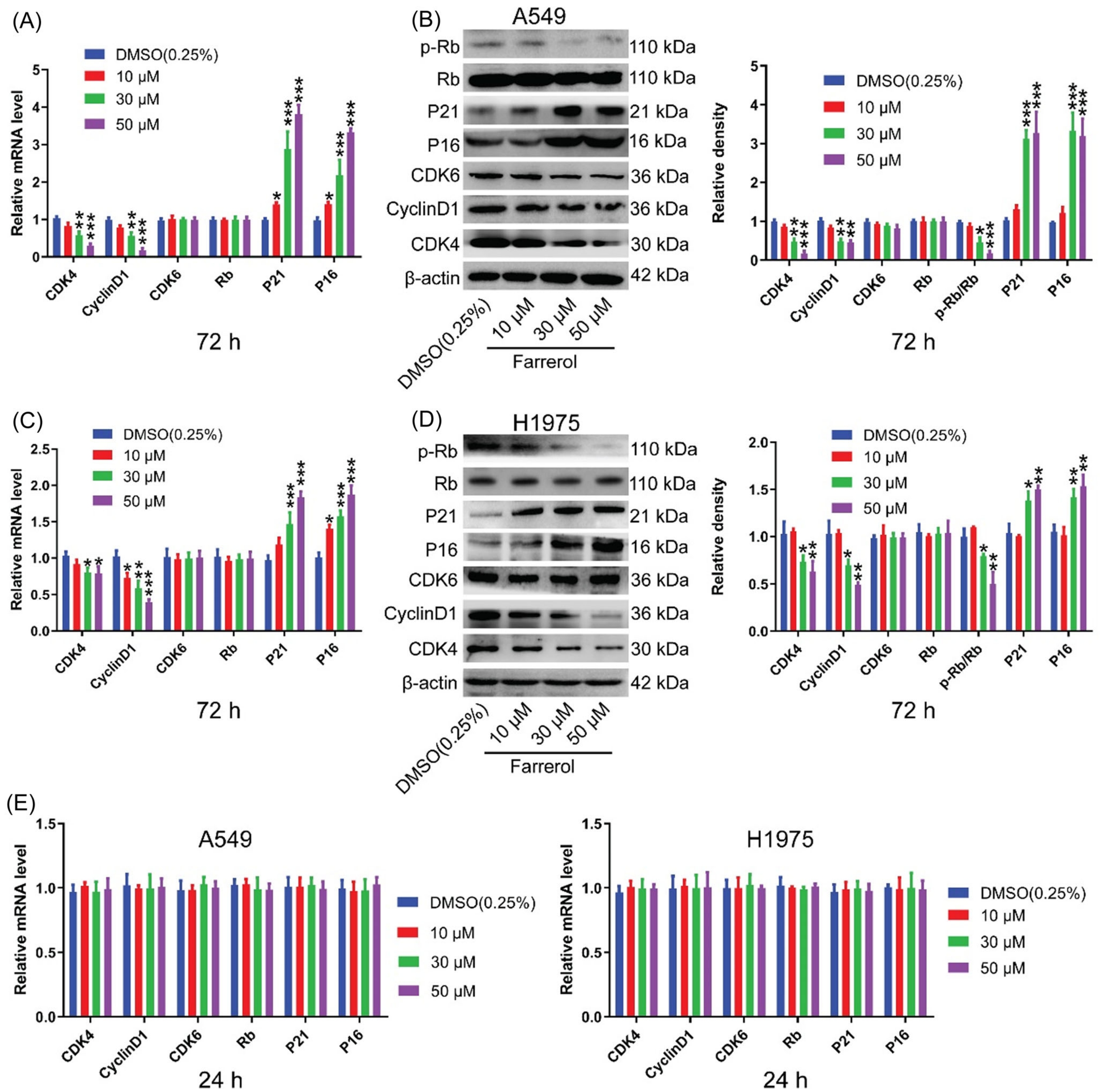

FIGURE 3 Effect of farrerol on the expression of cell cycle-related genes in lung adenocarcinoma cells. (A) Effects of farrerol treatment for 72 h on the mRNA levels of cell cycle-associated genes were measured by qPCR analysis in A549 cells. (B) Western blot analysis of protein extracts of A549 cells treated with different concentrations of farrerol for 72 h using antibodies against phospho-Rb, Rb, P21, P16, CDK6, CDK4, CyclinD1 or β-actin. Each lane contains 30 μg protein. (C) qPCR assay was performed to test the effect of farrerol treatment for 72 h on the mRNA levels of cell cycle-related genes in H1975 cells. (D) Western blot analysis was conducted to test the effect of farrerol treatment for 72 h on the protein expression of cell cycle-associated genes in H1975 cells. Left panel: representative immunoblots per group; right panel: statistical analysis. (E) Effects of farrerol treatment for 24 h on the mRNA expression of cycle-related genes in A549 (left) and H1975 (right) cells. Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, when compared with the control group. mRNA, messenger RNA; qPCR, quantitative polymerase chain reaction

4 | DISCUSSION

LC has the highest morbidity and mortality in cancer-related diseases globally.[1] Chemotherapy is still the front-line treatment for LC. Chemotherapeutic agents for LC, represented by platinum-based drugs, mainly produce antitumor effects by inducing DNA damage, resulting in cell cycle arrest and apoptosis.[29,30] Cell cycle progression is orchestrated by cyclins and cyclin dependent kinases (CDKs).[31] When cells are in the G0/G1 phase, Cyclin D1 is synthesized in large quantities and binds to CDK4 and CDK6

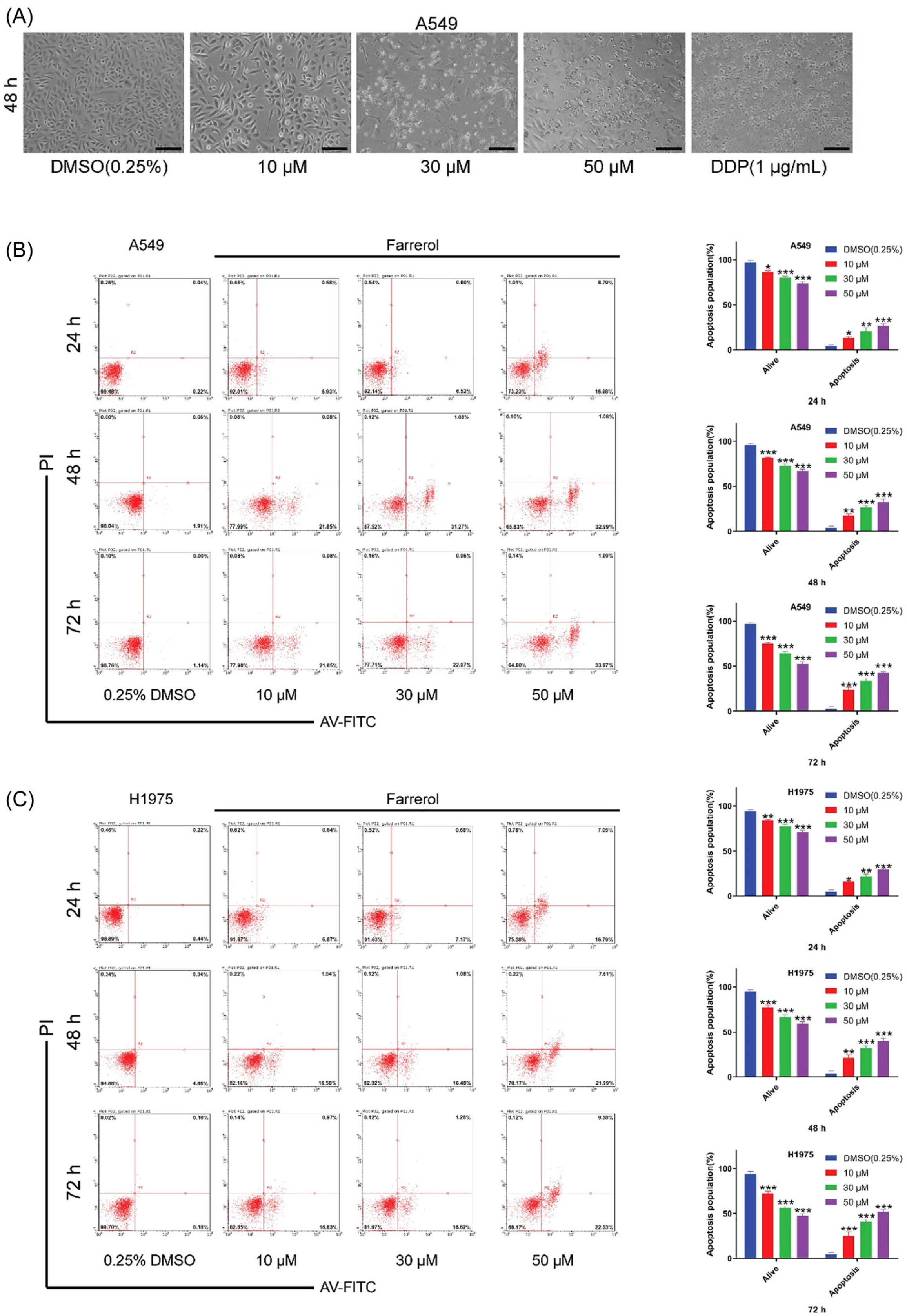

FIGURE 4 (See caption on next page)

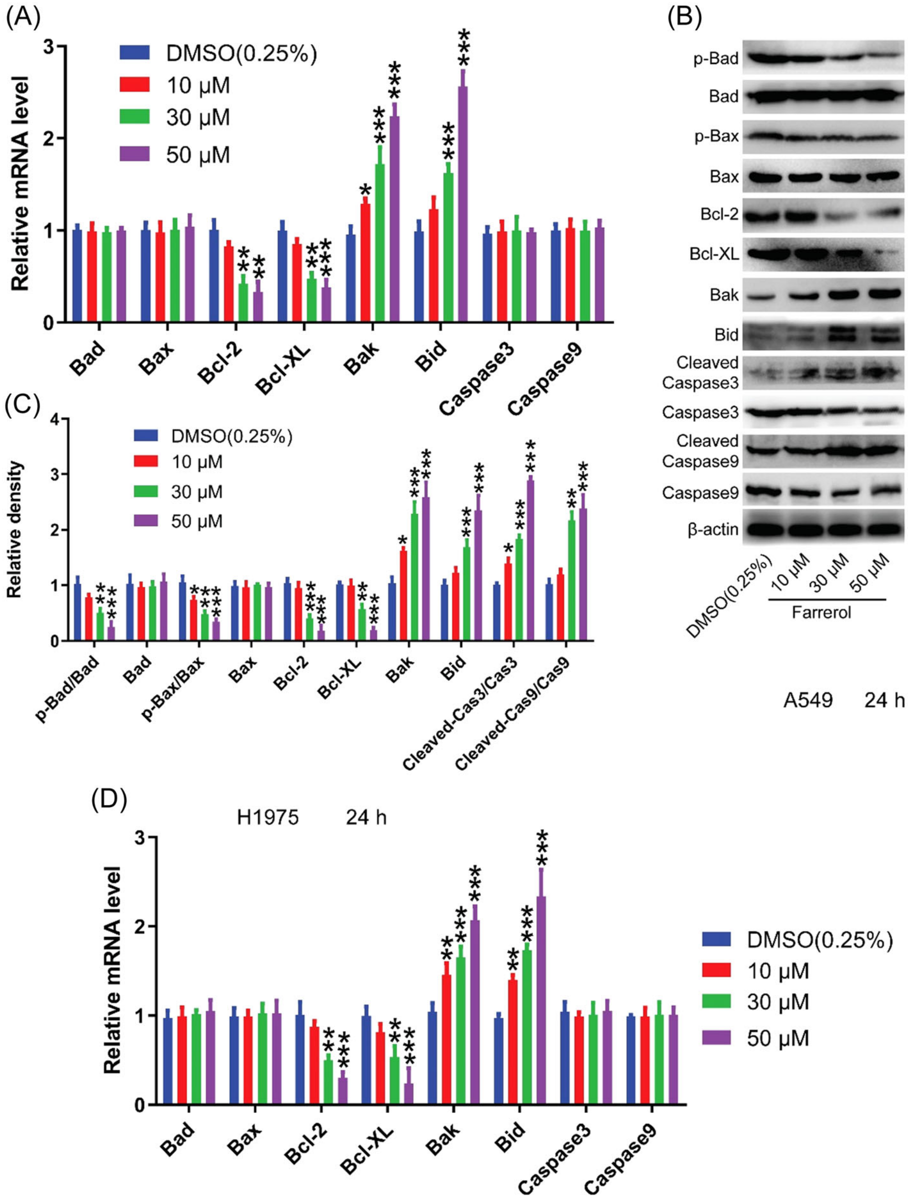

FIGURE 5 Farrerol regulates the expression of apoptosis-related genes. (A) Effect of farrerol treatment for 24 h on the mRNA levels of apoptosis-associated genes was tested by qPCR analysis in A549 cells. (B) Western blot analysis of protein extracts of A549 cells treated with farrerol for 24 h using antibodies against phospho-Bad, Bad, phospho-Bax, Bax, Bcl-2, Bcl-XL, Bak, Bid, cleaved-caspase3, cleaved-caspase9 or β-actin. Each lane contains 30 μg protein; (C) Quantification. (D) qPCR analysis was performed to test the effect of farrerol treatment for 24 h on the mRNA levels of cell apoptosis-related genes in H1975 cells. Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. **p* < 0.05, ***p* < 0.01, ****p* < 0.001, when compared with the control group. mRNA, messenger RNA; qPCR, quantitative polymerase chain reaction

FIGURE 4 Farrerol promotes apoptosis of lung adenocarcinoma cells in time-and dose- dependent manners. (A) Cell fragmentation and cell shrinkage after exposure to farrerol. Scale bar, 50 μm. (B) Representative flow cytometric analysis of apoptosis and necrosis of A549 cells treated with farrerol for 24 h (top), 48 h (middle) and 72 h (bottom) based on Annexin V (x-axis) and PI (y-axis) staining. (C) Flow cytometry with annexin V-FITC/PI double staining was used to determine apoptosis of H1975 cells treated with farrerol for 24 h (top), 48 h (middle) and 72 h (bottom). Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis for significant differences. Error bars represent the standard deviation of three independent experiments. **p* < 0.05, ***p* < 0.01, ****p* < 0.001, when compared with the control group. AV, annexin V; PI, propidium iodide

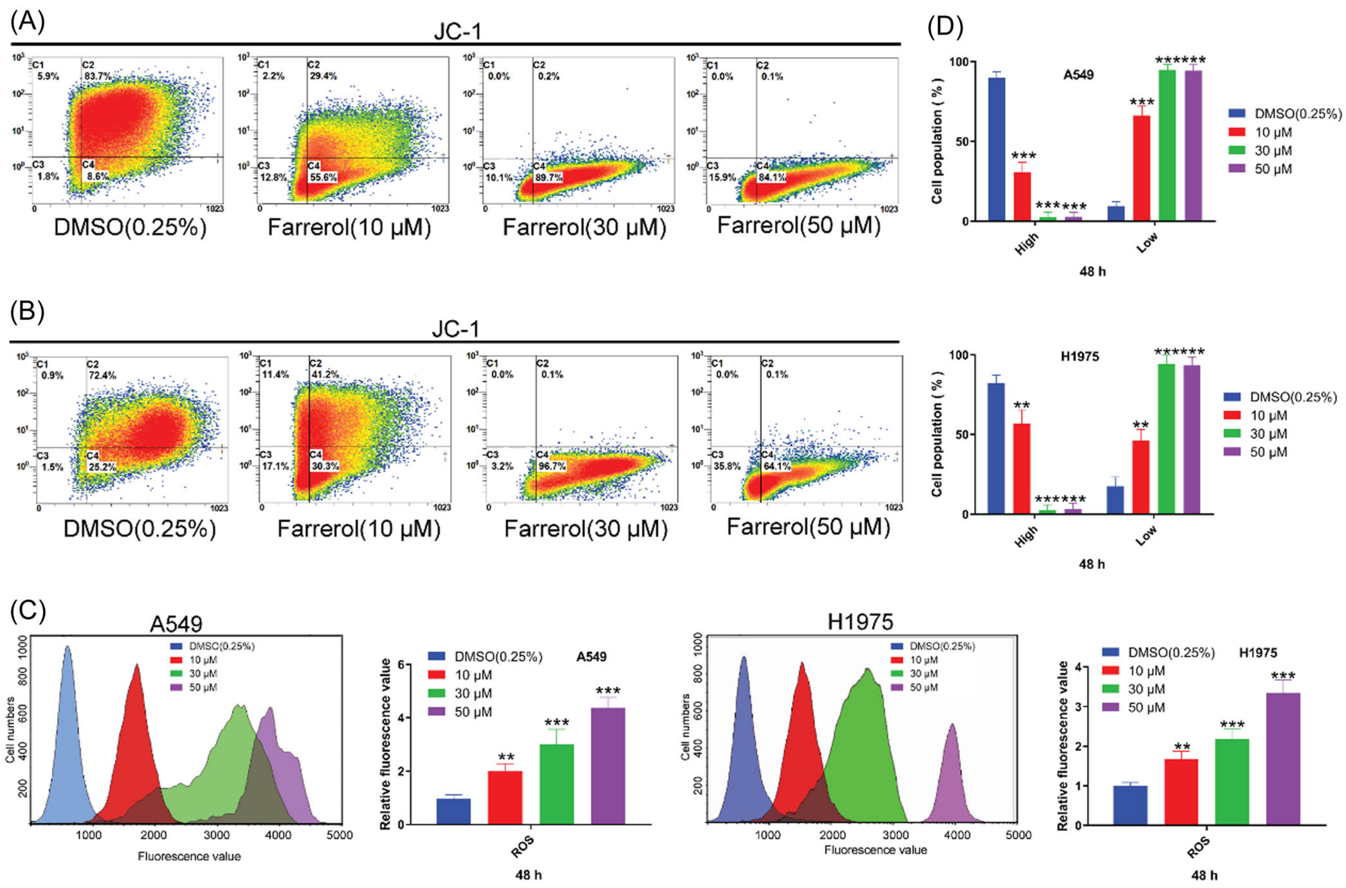

FIGURE 6 Farrerol treatment leads to increased ROS levels and decreased mitochondrial membrane potential in lung adenocarcinoma cells. (A, B) Effects of farrerol treatment for 48 h on the mitochondrial membrane potential of (A) A549 and (B) H1975 cells. Mitochondrial membrane potential was evaluated using JC-1 staining. (C, D) Effects of farrerol treatment for 48 h on ROS levels in (C) A549 and (D) H1975 cells. ROS level was measured by flow cytometry. Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. $^{**}p < 0.01$, $^{***}p < 0.001$, when compared with the control group. ROS, reactive oxygen species

specifically, promoting Rb phosphorylation and the transition from G1 to S phase.[32,33] Previous studies have identified dysregulation of cell-cycle checkpoints (Cyclin D1/CDK4/CDK6) in LC.[34,35] Our findings have verified that farrerol downregulated the expression of cell cycle-associated proteins, thereby blocking cell cycle progression. Furthermore, ROS affects cell cycle progression in a context-dependent manner through phosphorylation of CDKs and cell cycle regulatory proteins.[36] In the current study, we found that farrerol treatment resulted in a significant increase in ROS levels. The elevated ROS production from mitochondrial complex I in turn imparts increased amount of oxidative stress to cells, promoting cell death.[37] Nevertheless, effects of farrerol on cell cycle is insufficient to explain its inhibitory activity on proliferation of lung adenocarcinoma cells.

Effective induction of cell apoptosis is a common anticancer mechanism of drugs.[38] Apoptosis is a physiological controlled process regulated by complex signaling network,[39] of which mitochondrial apoptotic pathway is characterized by proapoptotic molecules released from mitochondria, activating caspase-9 and caspase-3, initiating the cell death cascade.[40] And the decrease in mitochondrial membrane potential is a critical event in the apoptotic cascade.[41,42] In our study, an increase in intracellular ROS levels and a decrease in mitochondrial membrane potential were observed in LAC cells exposed to farrerol. ROS inhibitor NAC resulted in a partial restoration of mitochondrial membrane potential in farrerol-treated LAC cells, suggesting that ROS was the cause of decreased mitochondrial membrane potential after farrerol treatment. The decline in mitochondrial membrane potential was followed by an increase in permeability of the mitochondrial membrane, giving rise to the release of apoptosis-inducing factors from the membrane into the cytoplasm and triggering the downstream apoptotic cascade.[43,44] Besides, our data indicated that Bax and Bad phosphorylation were reduced after farrerol treatment, contributing to their entry into the mitochondrial membrane and disrupting the mitochondrial membrane's integrity. Therefore, farrerol regulated mitochondrial apoptotic pathway in lung adenocarcinoma cells, facilitating cell apoptosis. It might be the primary reason for its inhibitory effects on lung adenocarcinoma cell proliferation. Likewise, in vivo experiments verified our findings in cellular experiments.

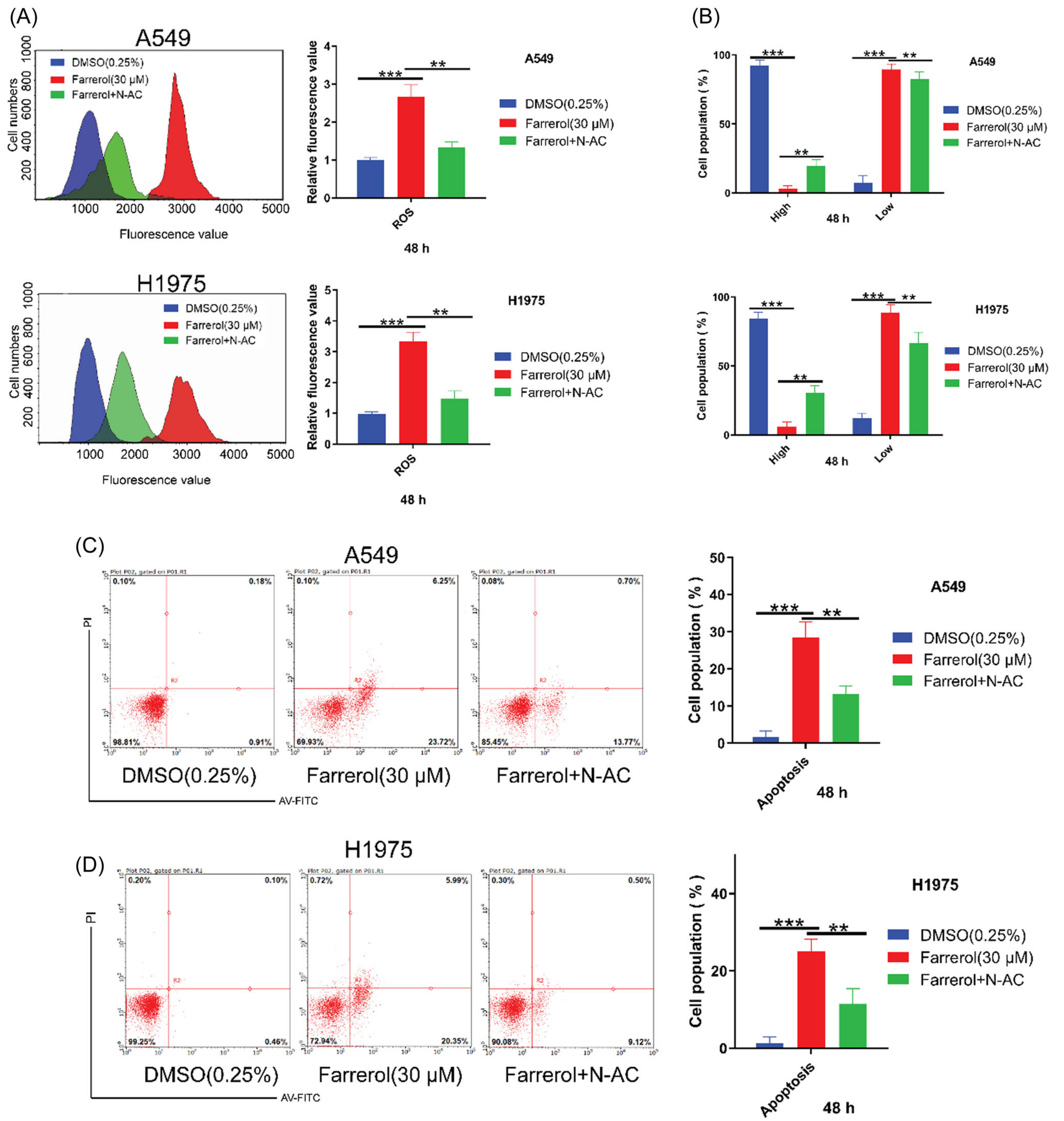

FIGURE 7 Effect of Farrerol combined with NAC treatment on ROS, mitochondrial membrane potential and apoptosis of lung adenocarcinoma cells. (A) ROS levels in A549 (upper panel) and H1975 (lower panel) cells treated with farrerol or in combination with NAC for 48 h. (B) Mitochondrial membrane potential in A549 (upper panel) and H1975 (lower panel) cells treated with farrerol or in combination with NAC for 48 h. (C, D) Effects of farrerol or/and NAC treatment for 48 h on apoptosis of (C) A549 and (D) H1975 cells. Left panel: representative images; right panel: statistical analysis. Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. ***p* < 0.01, ****p* <0.001, when compared with the control group. AV, annexin V; NAC, N-acetylcysteine; PI, propidium iodide; ROS, reactive oxygen species

Consistent with our experimental results, farrerol exercises antitumor function in laryngeal squamous cell carcinoma (LSCC) and gastric cancer (GC) via the mitochondrial-mediated pathway, where farrerol promoted apoptosis, upregulated cleaved caspase-3/9 levels, and downregulated Cyclin D1 levels.[45,46] Additionally, works in human ovarian epithelial cancer and GC showed that farrerol caused G0/G1-phase cell-cycle arrest and apoptosis through activating ERK signaling.[47,48] Farrerol inhibits the proliferation of multiple

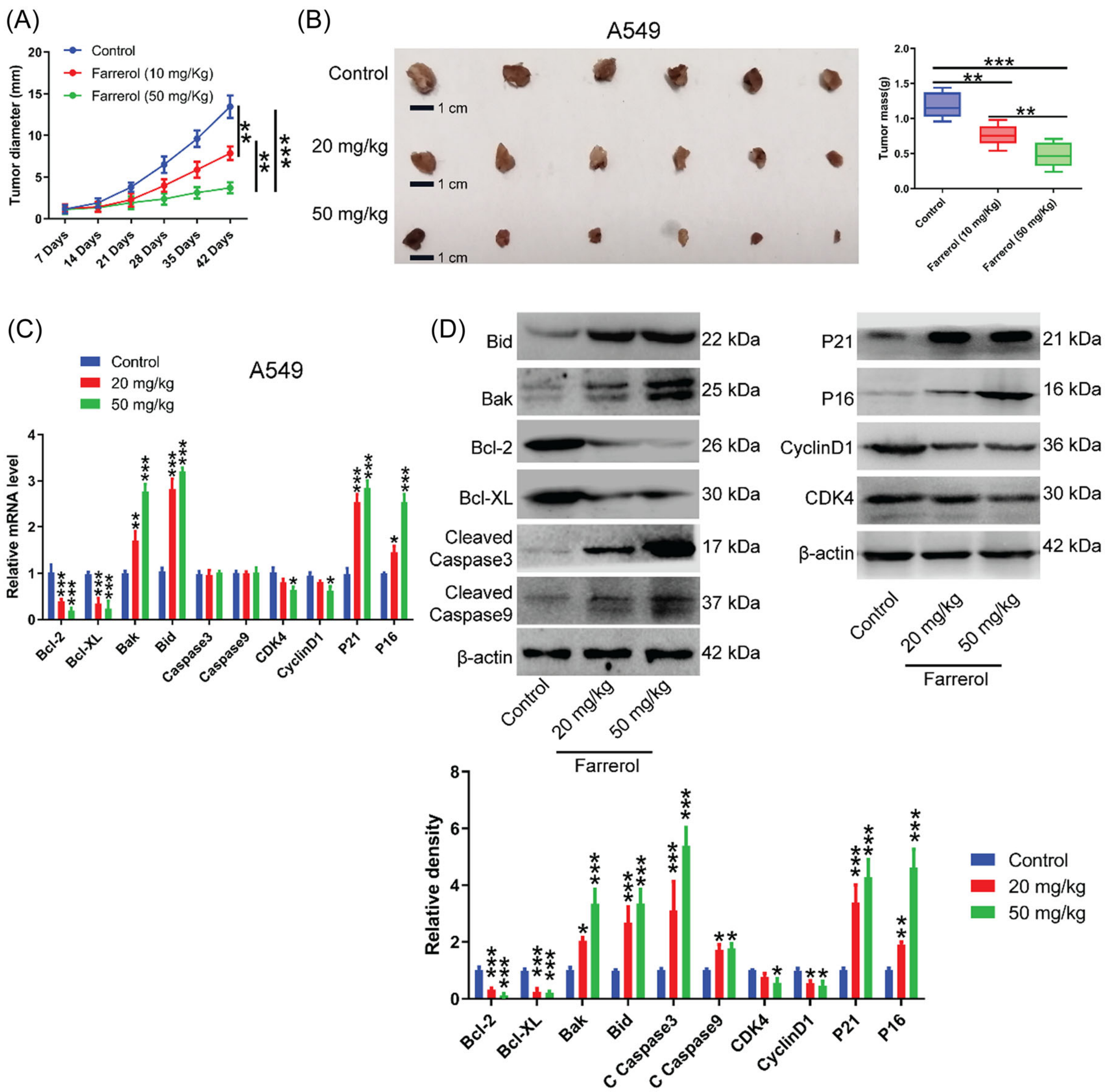

FIGURE 8 Administration of farrerol suppresses xenograft tumor growth. (A) Effect of farrerol administration on tumor diameter in A549 xenograft model. (B) Representative images of subcutaneous xenograft tumors (left) and quantification (right), 42 days postfarrerol administration. $N = 6$ mice per group. (C) Effects of farrerol administration on the mRNA levels of cell cycle- and apoptosis-related genes in tumor tissues ($n = 3$). (D) Western blot analysis was performed to test protein levels of cell cycle- and apoptosis-related genes in tumor tissues after farrerol administration for 42 days ($n = 3$). Data were analyzed by one-way analysis of variance (ANOVA) with Dunnett posttest analysis. Error bars represent the standard deviation of three independent experiments. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, when compared with the control group. mRNA, messenger RNA

different cancer types by regulating various signaling pathways. It indicates that farrerol can be applied to treatment of multiple tumors.

Regarding the dosages used, Zhang et al.[45] have investigated the effect of Farrerol on LSCC cancer and uncovered inhibitory effect of farrerol (10, 20, 50 μM) on LSCC cells. Besides, they found that 40 mg/kg of farrerol treatment by intraperitoneal injection significantly decreased tumor volumes and weight. In this study, high concentrations of ferrerol (10, 30, and 50 μM) were also used in in vitro experiments based on the IC_{50} value (18.9 μM) of farrerol. And in nude mice inoculated with A549 cells, the mice were administrated with 20 or 50 mg/kg of farrerol by gavage. Although ferrerol significantly inhibited tumor growth, this study did not report on toxic effects to organs including heart, liver, and kidney in the presence of farrerol during experiments. Previously, Ma et al.[49] have explored farrerol activity on cisplatin-induced chronic kidney disease, in which body weight of C57BL/6 mice that were administered farrerol at 10 mg/kg by intraperitoneal injection were measured at 0, 3, 7, 14, 21, 28, and 38 days. No change in body weight, kidney index, BUN, and SCr in farrerol-treated mice was found compared to untreated control; moreover, hematoxylin and eosin-staining of kidney sections showed that there was no damage to renal tubular both in farrerol-treated and untreated groups. Farrerol suppressed the release of proinflammatory factors, protecting kidney function. Similar results were obtained by other authors.[50] In addition, a recent study showed that oral gavages of farrerol (45 mg/kg/day) did not affect body weight, colon length, the normal mucosal architecture of C57BL/6 mice compared with the no treatment group.[51] These evidence suggest that farrerol has no toxic side effects on mice. Nevertheless, there is an apparent limitation that drugs at current experimental working concentrations cannot directly translate to the clinic. Therefore, structural modification of natural products is needed to overcome existing shortcomings and to increase its potency and activity.[52,53]

5 | CONCLUSIONS

In summary, we report here that farrerol effectively restrains LAC cell proliferation via regulating the expression of key proteins involved in the transition from G1 to S phase. Farrerol treatment causes G0/G1 cell cycle arrest and activates the mitochondrial apoptotic pathway by elevating ROS levels, inducing cell apoptosis. Farrerol exhibits antitumor effect on lung adenocarcinoma progression. These findings provide experimental evidence for the potential application of farrerol in treatment of lung adenocarcinoma.

AUTHOR CONTRIBUTIONS

Chuanshu Cai: study conceived and designed. **Yi Guo and Quan Li**: performed experiments. **Chuanshu Cai and Rongmu Xia**: participated in data analysis. **Yi Guo**: wrote the manuscript. **Quan Li, Rongmu Xia, and Chuanshu Cai**: revised the manuscript. All authors approved the final version of the manuscript.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this published article and its supplementary information files.

ETHICS STATEMENT

Animal experiments in this study were approved by the Ethics Committee of The First Affiliated Hospital of Fujian Medical University. Animal experimental procedures strictly followed the Guidelines of the China Animal Welfare Legislation for Care and Use of Laboratory Animals.

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